

Difference	DFA	NFA
Multiple transitions	1 exiting arrow	≥ 0 exiting arrows
Epsilon transitions	$\times$	$\checkmark$
Missing transitions	No missing transitions	Missing transitions mean transitions to sink/reject state

- Nondeterministic = Event paths cannot be determined precisely
- Finite = Finite and small amount of space used
- Automaton = Computing machine

Definition

A **nondeterministic finite automaton (NFA)** M is a 5-tuple

$M = (Q, \Sigma, \delta, q_0, F)$, where,

1. Q : A finite set (**set of states**). $\triangleright$ **Space (computer memory)**
2. Σ : A finite set (**alphabet**).
3. $\delta : Q \times (\Sigma \cup \epsilon) \rightarrow P(Q)$ is the **transition function**, where $P(Q)$ is the power set of Q . $\triangleright$ **Time (computation)**
4. q_0 : The **start state** (belongs to Q).
5. F : The set of **accepting/final states**, where $F \in Q$.

Uses of NFA's

- Constructing NFA's is easier than directly constructing DFA's for many problems.
Hence, construct NFA's and then convert them to DFA's.
- NFA's are easier to understand than DFA's.

Lecture 6

An NFA can be thought of as a DFA that can be in many states at once.

At each point in time, when the NFA needs to follow a transition, it tries all the options at the same time.

The NFA accepts if any of the states that are active at the end are accepting states. It rejects otherwise.

Any language that can be accepted by a DFA can be accepted by an NFA.

Why?

- Every DFA essentially already is an NFA!

Question: Can any language accepted by an NFA also be accepted by a DFA?

Surprisingly, the answer is yes!

Type 1: No epsilon involved in this NFA.

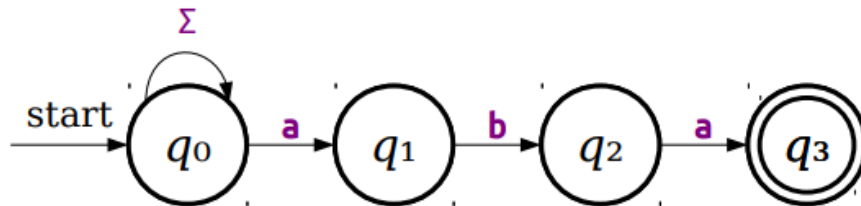
Note: $\Sigma = a, b$

Since we want to make a DFA, we know that for every character in the alphabet there should be a transition for each state. There can be only one transition in a state for each character in the alphabet.

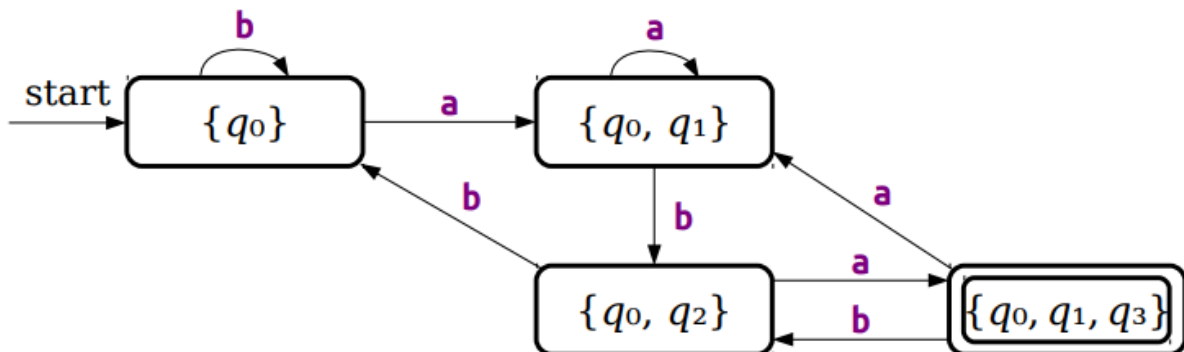
Draw a DFA table, with two columns a and b, since the alphabet contains 2 characters.

Start from the starting state.

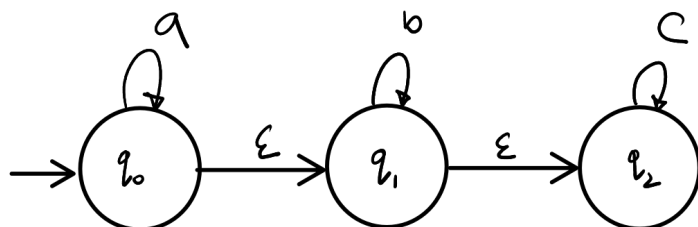
- Note down the states from which $\{q_0\}$ goes to by reading a from the NFA.
- Do the same thing for b.
- We get $\{q_0, q_1\}$ by reading a. This is a new state.
- So in the next row, check what each state in the set $\{q_0, q_1\}$ go to a and b and note down the new set of states.
- When do we stop? We stop when we don't get any new set of states by reading a and b.
- What is the final state, the final state in the NFA is q_3 , notice where this state is included in the set of the states. That becomes the final state of DFA.



	a	b
$\{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$\{q_0, q_2\}$	$\{q_0, q_1, q_3\}$	$\{q_0\}$
$*\{q_0, q_1, q_3\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$



Type 2: Epsilon involved.



① Draw NFA table

	a	b	c	ϵ
q_0	$\{q_0\}$	$\emptyset$	$\emptyset$	$\{q_1, q_2\}$
q_1	$\emptyset$	$\{q_1\}$	$\emptyset$	$\{q_2\}$
q_2	$\emptyset$	$\emptyset$	$\{q_2\}$	$\emptyset$

ϵ -Closure: When we take ϵ -closure of a state the state itself is included.

ϵ -Closure (q_0) = $\{q_0, q_1, q_2\}$ Note down where q_0 goes to by reading ϵ - from table.

ϵ -Closure (q_1) = $\{q_1, q_2\}$

ϵ -Closure (q_2) = $\{q_2\}$

When we take epsilon Closure of start state, the new state we get becomes the start state.

Note; When there is an ϵ outgoing from start state, the start changes.

② Draw DFA table.

	a	b	c
$A = \{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_2\}$
$B = \{q_1, q_2\}$	$\emptyset$	$\{q_1, q_2\}$	$\emptyset$
$C = \{q_2\}$	$\emptyset$	$\emptyset$	$\{q_2\}$
Dead = $\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

$\{q_0, q_1, q_2\}$

lets analyse what happens in starting state when reading

- Check where q_0 goes to a by reading a, it goes to q_0 .

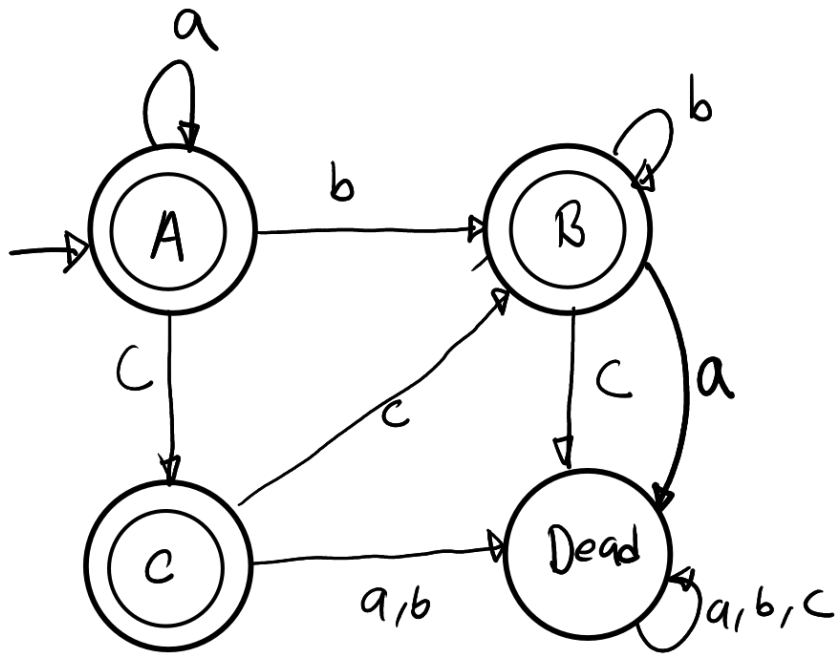
Take ϵ -closure of q_0 and note it down in table.

ϵ -Closure (q_0) = $\{q_0, q_1, q_2\}$

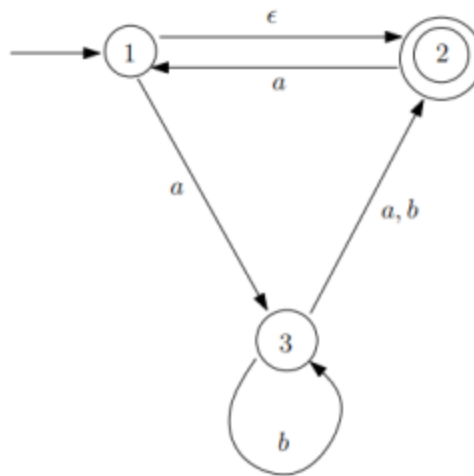
- Now Check where q_1 and q_2 goes to by reading a. It doesn't go anywhere. Therefore, the by reading a $\{q_0, q_1, q_2\}$ goes to

$\{q_0, q_1, q_2\}$

Lecture 6



Convert the NFA to a DFA.



- NFA M is specified as
 Set of states is $Q = \{1, 2, 3\}$
 Set of symbols is $\Sigma = \{a, b\}$
 Start state is 1
 Set of accept states is $F = \{2\}$
 Transition function δ is:

δ	a	b	ϵ
1	$\{3\}$	ϕ	$\{2\}$
2	$\{1\}$	ϕ	ϕ
3	$\{2\}$	$\{2, 3\}$	ϕ

- How do you convert this NFA to DFA?

This was solved in class.

Downside:

- In converting an NFA to a DFA, the DFA's states correspond to sets of NFA states.
- **Useful fact:** $|\wp(S)| = 2^{|S|}$ for any finite set S .
- In the worst-case, the construction can result in a DFA that is *exponentially larger* than the original NFA.